Aashan Javed

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Research Interests

Calibrated uncertainty for frozen foundation models under distribution shift, applied to climate and urban heat. My work asks whether published reliability guarantees survive the shift a deployed model actually meets, and so far the answer has been no.

Education

National University of Computer and Emerging Sciences (FAST-NUCES) 2020 – 2024
Bachelor of Science in Computer Science *Islamabad, Pakistan*

- Awarded the **Gold Medal** and a place on the **Rector's List of Honour** for the **seventh semester**. Batch rank **57 of 285**, the **top 20 percent**.

Preprints

When Does a Frozen Time-Series Foundation Model's Uncertainty Fail, and What Actually Fixes It?
A. Javed
Under review, Foundation Models for Time Series workshop, NeurIPS 2026 (non-archival). code
A frozen forecaster's own prediction intervals fall furthest short where the series has **drifted most**, and the failure tracks **regime shift rather than forecast horizon**. Three static post-hoc conformal variants do not repair that conditional gap; running conformal **online narrows it without closing it**. The natural rolling-origin implementation **leaks its own future into calibration** and overstates the recovery, so every result here lags feedback by the full horizon.

Research Experience

Independent Researcher 2025 – Present
Calibrated uncertainty under distribution shift *Remote*

- Designed and ran **the experiments behind my research end to end**: problem selection, literature gap verification, data acquisition, implementation, evaluation and writing.
- Built a **model-agnostic, unit-tested conformal prediction library** (split-conformal, CQR, Mondrian, adaptive conformal inference) reused across every study.
- Ran an **adversarial audit of my own results** and found that a masking flag I believed was off had never reached the frozen encoder, so two calls on the same input disagreed. **I stopped the affected work rather than publish on it.** Reproducing a frozen encoder's output before trusting it, and reporting a falsified prediction rather than discarding it, are now fixed steps in how I work.

Professional Experience

Founder and CEO Jan 2024 – Present
Valfirst AI (Valfirst Software (SMC-Private) Ltd), side venture *Pakistan / Remote*

- Founded and lead **Valfirst AI**, which builds and operates **autonomous AI agents** that run real business workflows end to end for operations-heavy small and mid-sized companies, not chatbots but digital workers that respond to leads, qualify customers, and handle repetitive processes.
- Built and deployed **11 live, interactive AI agents** across industries, demoable with **no login** at valfirstai.com; agents respond to and qualify a lead in **seconds**, integrate into a client's tools, and report via a monthly scorecard.
- In pilot: running the **first paid project**, an agent system that manages the **digital media operations for Proton Studios**.

Machine Learning Engineer (R&D) Jan 2025 – Present
FortyGuard, urban heat and climate intelligence *Abu Dhabi, UAE (Remote)*

- Built the **Large Temperature Model (LTM)**, forecasting **street-level (2 m) air temperature at 10 m resolution** across the **United States**.
- Led a collaboration with **NVIDIA** on a custom **CorrDiff diffusion-based downscaling model**, a **255× spatial amplification** (16× finer per axis) taking a 25 km grid to roughly 1.5 km, for high-fidelity urban climate modelling (San Jose region).
- Currently building **FortyGuard's geospatial agents**. Own **multi-source geospatial pipelines** (**10M+ observations/day**) and **model evaluation and drift monitoring in production**, where the gap between reported accuracy and deployed reliability became concrete.

Software Engineering Intern Jun 2024 – Dec 2024
FortyGuard *Abu Dhabi, UAE (Remote)*

- Built **dashboards** over **100K+ records/day** and the data plumbing the temperature modelling later ran on.

Machine Learning Engineer Intern Jun 2023 – Sep 2023
Payactiv *Islamabad, Pakistan*

- Built **gradient-boosted risk scoring** over **1M+ transaction records**, where **calibrated abstention** mattered more than raw accuracy.

Community & Service

FortyGuard Global AI Hackathon '26 2026
Mentor *Remote*

- Served as a **mentor** for a global climate-tech hackathon and led the session **"Build with Heat Intelligence"** on 19 August 2026, on working with temperature and heat data in applied projects.

Climate Change AI (CCAI) 2025 – Present
Community member and volunteer *Remote*

- Active member of the **Climate Change AI** community; collaborate on and volunteer at its research events.

Alkhidmat Foundation Pakistan 2022
Digital Volunteer, Flood Relief *Pakistan (Remote)*

- Supported digital outreach and awareness during the **2022 flood emergency**, amplifying relief and donation drives to connect support with affected communities. Recognised with a Certificate of Appreciation.

FAST Community Service Society 2021 – 2023
Head of Information and Communication *Islamabad, Pakistan*
- Led communications and volunteer coordination for FAST's student-run community service society, running its social media, announcements, and event promotion, and mobilising volunteers across charity drives and student outreach activities.

Google Developer Groups (GDG) Islamabad 2022 – 2024
Community Volunteer *Islamabad, Pakistan*
- Helped plan and run developer meetups, workshops, and technical events, supporting participant engagement and connecting students with the local tech community.

Youth For Pakistan 2021 – 2024
Campus Ambassador *Islamabad, Pakistan*
- Represented Youth For Pakistan on campus for over two years, promoting youth engagement and organising student outreach for awareness campaigns and volunteer opportunities.

Student Mentoring 2021 – Present
Peer Mentor *Pakistan*
- Mentor students on university admissions, degree and field selection, and how to navigate early studies, both during my time at FAST and since.

Technical Skills

Uncertainty & Evaluation: **Conformal Prediction** (split, CQR, Mondrian, adaptive), Calibration Assessment, Coverage & Interval-Width Analysis, Distribution-Shift Evaluation, Leakage-Controlled Experimental Design
Machine Learning: **Python, PyTorch, scikit-learn, XGBoost**, Foundation-Model Representation Probing, Time-Series Forecasting, Spatiotemporal Modelling
Geospatial: Sentinel-2, Landsat C2, Microsoft Planetary Computer, rasterio, Cloud-Optimised GeoTIFF, Local Climate Zones
Research Engineering: Reproducible Pipelines, **Docker, CI/CD**, LaTeX, Git. **Languages**: English (**IELTS Academic 7.5**), Urdu (native).